Federated Multiaccess Edge Computing Architecture for Scalable and Secure U-Space Traffic Management in 5G-Enabled UAS Networks

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***Abstract*—This paper addresses the urgent need for practical telecommunication standard-based architectures to meet the growing demand for reliable, low-latency 5G communication in U-space, where existing frameworks struggle with dense, multi-operator UAV traffic. We propose a federated data mesh architecture leveraging Multiaccess Edge Computing (MEC) integrated with 5G, enabling scalable, secure, and interoperable traffic management across decentralised and centralised MEC deployments. Simulations comparing centralised cloud-only, USSP-centric MEC, and CISP-centric MEC models demonstrate improved latency and opersational feasibility. The study also explores critical challenges such as inter-MEC communication, data federation, security, privacy, and economic trade-offs, providing essential insights for designing resilient, efficient UAS traffic management infrastructures to support future Advanced Air Mobility (AAM) ecosystems.**

***Keywords*—U-Space, 5G, Multiaccess Edge Computing (MEC), Data Mesh, Very Low Level (VLL) airspace**

I. Introduction

The rapid expansion of Unmanned Aerial Systems (UAS) has fundamentally transformed modern aviation, enabling novel applications in commercial, emergency response, and urban air mobility domains. Advanced Air Mobility (AAM) initiatives [1], involving autonomous aerial vehicles and electric vertical takeoff and landing (eVTOL) aircraft [2], are poised to revolutionize transportation by fostering sustainable, safe, and rapid connections beyond conventional ground infrastructure limitations. These advancements, however, require new operational paradigms, particularly in Very Low Level (VLL) airspace, to address the complexity and scale of UAS deployments and integrate effectively into emerging AAM ecosystem.

The aviation sector has responded to the increasing complexity of low-altitude operations by adopting next-generation technologies such as 5G and Wi-Fi communications, LiDAR and vision-based navigation, and digital-twin surveillance [3], all of which are reshaping Communication, Navigation, and Surveillance (CNS) architectures for dense urban and VLL environments [4], [5]. Existing airspace management systems, devised for manned aviation, are becoming inadequate for the demands of dense UAS deployments, which has stimulated global efforts toward robust Unmanned Traffic Management (UTM) frameworks [6]. Leading initiatives such as the FAA's UTM [7] in the United States and the European U-space [8] leverage automation and digitalisation to ensure the safe and scalable coexistence of autonomous and traditional airspace users. Among these technological advances, the integration of 5G mobile networks stands out as a critical enabler for UAS traffic management, providing ultra-reliable, low-latency connectivity that supports real-time command, control, and safety functions essential for autonomous UAS operations in regulated airspace [9], [10].

Building upon this foundation, mobile network infrastructure, particularly Multiaccess Edge Computing (MEC) integrated with 5G emerges as a pivotal enabler for the real-time positioning, and traffic management required for advanced UAS operations. While these telecommunication platforms offer significant technical promise, their adoption presents substantial challenges. Aligning U-space rollout strategies with evolving 3GPP [11], and ETSI - telecommunication standards and infrastructure [12], harmonizing the interests of diverse stakeholders, and establishing scalable, federated governance models remain ongoing concerns. In response, this paper introduces and evaluates a novel MEC-enabled, data-mesh-driven architecture for U-space traffic management, emphasizing service migration, latency constraints, and domain-specific data ownership, and presents practical insights through a simulation case study.

II. Background

Here we present standards and exploratory research from literature on U-Space, UAS traffic management and 5G technology that serve as an understanding to build the federated integration framework with data mesh principles and 5G-MEC services.

A. U-Space and its Services

U-Space is Europe's framework for digitalised UAS traffic management that enables safe, scalable and automated operations in VLL airspace. It defines progressive service levels (U1–U4) that span e-registration and flight planning through to tactical deconfliction and collaborative separation, and it is implemented through a set of regulatory acts, implementing rules and guidance produced by the EU and EASA [13], [14]. The U-Space vision and roadmap have been formalised in SESAR work products and the U-Space Concept of Operations, which set out the required services, interfaces and phased deployment approach for aviation authorities, Common Information Service Providers (CISPs), U-Space Service Providers (USSPs) and airspace users.

Operationally, UTM/U-Space systems provide a mix of strategic services (flight authorization, NOTAM/constraints distribution, logging) and tactical services (real-time telemetry, conflict detection, dynamic geo-fencing) [15]. These services demand different network properties: strategic functions tolerate higher latency and centralised processing, while tactical functions require low, bounded end-to-end latency, high availability and deterministic continuity during handovers [16]. The European policy and standards ecosystem (including EASA guidance, SESAR (Single European Sky ATM Research 3 Joint Undertaking) projects and EU standardisation coordination) emphasises interoperable data models, common interfaces and certified CISPs/USSPs to assure safety and regulatory compliance across member states [15], [17].

Several large-scale European projects have explored these themes. EURODRONE [18] focuses on developing interoperable UTM/U-Space services for cross-border UAS operations, validating data exchange standards and harmonised interfaces across Member States. 5G!Drones [19], as reported by Gracia et al., demonstrated how 5G connectivity, slicing, and edge computing can enhance UAS command-and-control and payload use cases, confirming that telecom-grade features are critical enablers for U-Space. From our perspective, these projects show the clear advantages of coupling aviation service models with 5G infrastructure. However, they leave open critical questions around how to structure multi-tenant MEC for tactical U-Space services and how to enforce federated data governance (data mesh) across USSPs, CISPs, and MNOs; gaps that our work addresses by proposing and analysing a federated MEC data mesh architecture.

B. 5G as technology enabler for U-Space

The telecom foundation for U-Space rests on the 5G Service-Based Architecture (SBA) and the extension of cloud capabilities to the network edge (MEC). The 5G SBA (as specified in the 3GPP system architecture and procedures) decouples network functions into discoverable services and enables network slicing, QoS exposure and API-based capability sharing primitives that are essential for integrating UTM applications with mobile networks and for maintaining control-plane session continuity as UAS move across cells and operators.

MEC [20], [21] brings compute and platform services close to the RAN (Radio Access Network) or radio network so that latency-sensitive U-Space functions (tactical deconfliction, remote identification, rapid flight approvals) can execute near the radio. ETSI's MEC framework and federation work provide the reference architecture, APIs and operational guidance for deploying MEC at multiple layers: far edge (RAN-co-located), metro/PoP (Point of Presence) edge (regional aggregation) and central cloud, while MEC interoperability and federation specifications describe how state, applications and discovery can be shared across operator and edge domains [22], [23]. These standards, together with ITU IMT-2020 [24] guidance for radio performance, form the technical basis for designing U-Space deployments that meet stringent latency and availability targets [25].

Mobile Network Operators (MNOs) play a pivotal role: they provide the RAN, transport, data and control plane anchoring. They also host MEC PoPs; they implement network slices and QoS management needed for guaranteed tactical channels; and they participate in inter-RAN mobility and relocation that underpin user-plane continuity [26], [27]. For U-Space, practical deployment choices (far edge vs metro MEC vs central cloud) involve trade-offs between latency, manageability and cost where metro-edge MEC typically offers the best trade-off for tactical U-Space services by combining low latency with economies of scale and multi-USSP interoperability [28], [29]. ETSI MEC guidance and 3GPP session continuity mechanisms must be complemented by commercial agreements (MEC hosting, federation, SLAs) and regulatory oversight to ensure neutrality, security and reliability [23], [30].

Research Contribution: This paper introduces a novel MEC-enabled, data-mesh-driven architecture for U-Space traffic management that aligns with 3GPP 5G standards and ETSI MEC framework, addressing latency, scalability, and federated governance challenges.

- We propose a MEC-based U-Space architecture leveraging 3GPP service-based principles, supporting both centralised and decentralised deployments with domain-specific data ownership for USSP and CISP roles.
- We elaborate on the application of Data Mesh principles to U-Space, emphasizing federated governance, domain-oriented data product ownership, and event-driven information management for secure, real-time, and interoperable multi-actor operations.
- We evaluate latency and scalability constraints, focusing on MEC placement, migration, and edge–cloud orchestration, and present a simulation based study demonstrating key performance indicators for flight approval and tactical service responsiveness.

III. 5G-MEC enabled Federated data architecture in U-Space

Edge computing can be deployed at multiple scales - far edge, metro edge, and hyperscaler/central cloud layers. The far-edge MEC, often co-located with base stations or radio units, delivers ultra-low latency and immediate access to radio network data, but it suffers from limited compute capacity, high deployment cost, and complex lifecycle management. By contrast, the metro edge offers a *"sweet spot"* for many latency-sensitive applications [31]: close enough to end-users or UASs to provide round-trip times in the 10–20 ms range, while also benefiting from economies of scale, stronger connectivity, and more manageable operational overhead. Hyperscaler clouds, while powerful, are geographically distant and unsuitable for real-time tactical loops. From the perspective of U-Space, metro MEC emerges as the most balanced option: it offers sufficiently low latency for tactical decisioning, wide coverage across multiple 5G tiles, and infrastructure that can be governed, scaled, and federated across USSPs and CISPs more feasibly than relying on many far-edge nodes or centralised cloud-only instances. In this work, we therefore envision U-Space deployments where multiple base stations are geographically co-located, with MEC-enabled traffic management functions such as flight authorization, geo-awareness, and tactical deconfliction hosted at the metro edge.

Building on this, the design of U-Space planning must explicitly integrate MEC with 5G coverage tiles and involve mobile network operators (MNOs) in its operation. Two broad orientations are possible: a centralised model and a decentralised MEC-enabled model. In a purely centralised approach as demonstrated in figure 1, no MEC is deployed within the U-Space; both CISPs and USSPs instead rely solely on their respective cloud infrastructures, typically exposing services through standard APIs such as MQTT. The UAS with a 5G modem and an active subscription to the MNO, connects to the geographically located RAN or radio unit. The MNO's gateway allocate required resources in its channel represented as data and control link with relavant user and control plan functions - enabling connection to the USSP in the global network. While this approach provides a clean, cloud-native model, it introduces unavoidable latency overheads, since tactical loops remain remote and dependent on the transport core.

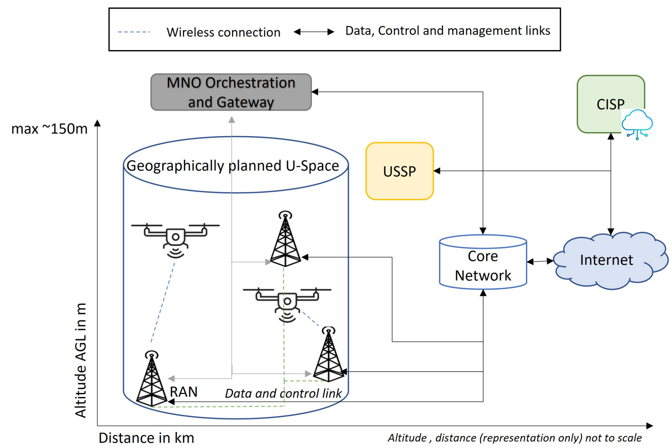

Figure 1. Communication infrastructure in a centralised model

By contrast, a decentralised MEC-enabled architecture leverages edge resources at metro or far-edge points of presence to host tactical decisioning closer to operations. This enables local, low-latency autonomy while still preserving central oversight. However, decentralization also raises critical questions of data ownership and governance - particularly given the multiplicity of stakeholders in U-Space.

Here, the concept of data mesh offers a promising paradigm. Rooted in domain-oriented data ownership, data mesh dis-

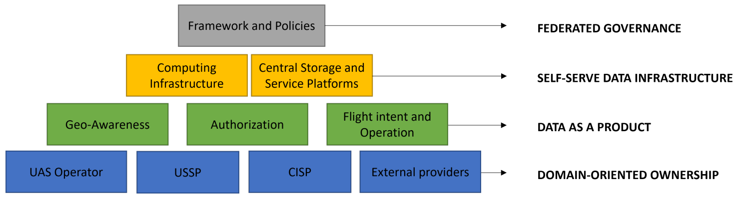

Figure 2. Representation of data mesh principle in U-space data ecosystem

tributes responsibility for data products across the relevant operational domains rather than concentrating it in a single monolithic infrastructure. In a U-Space setting, this means that each USSP or CISP domain manages its own data products such as telemetry streams, flight authorizations, or geo-awareness updates published through federated event buses with agreed schemas and SLAs. This structure as in figure 2 is essential in decentralised architectures, where tactical decision-making cannot rely solely on a central authority but must be locally actionable and shared across domains in near real-time. MEC plays a pivotal role in this vision: MNO-hosted or federated MEC platforms provide the computational substrate for hosting domain-specific microservices, while ensuring that cross-domain data products can be exchanged with ultra-low latency [32], [33].

The four principles of data mesh align naturally with U-Space:

- **Domain-oriented ownership** where UAS operators, USSPs, CISPs, external providers like weather data service, ATM and ANSPs each own their respective data products (mission/telemetry, flight authorization, regulatory constraints).
- **Data as a product** where each stakeholder publishes consumable feeds (e.g., "geo-awareness" or "flight intent") instead of raw streams.
- **Self-serve data infrastructure** where MEC and cloud provide shared, standardized publishing and subscription platforms.
- **Federated governance** ensuring common schemas, SLAs, and security policies across all participants. Together, these principles ensure interoperability without undermining local autonomy.

Within a decentralised MEC-enabled U-Space, two distinct deployment models can be identified. In the USSP-centric model in figure 3, each USSP may operate its own MEC application instances, either co-located with MNOs or via neutral hosts. At takeoff, UAS traffic is routed via the RAN to the USSP's MEC service instance (the yellow disk), where tactical loops and telemetry are processed locally. USSP access the application service on the MEC at a particular U-Space through service agreements with the MNO. And this MNO's orchestration and gateway manages connectivity and service application of the USSP in its platform, ensuring that traffic from UASs is properly routed either locally (to the MEC applications) or outward (to the Core Network and Internet) via control and data links. With the advancements in virtual network functions several applications of various USSPs similar to USSP A in figure 3 can be hosted. Furthermore, this fits the commercial logic of USSPs, allowing them to control operational data and monetize differentiated services. However, scalability is problematic: every USSP would need MEC deployments across all U-Spaces or negotiate with multiple MNOs, leading to high operational overhead and risks of fragmentation. Inconsistencies in MEC implementation across USSPs could jeopardize safety-critical tactical behavior. By contrast, the CISP-centric model in figure 4 -

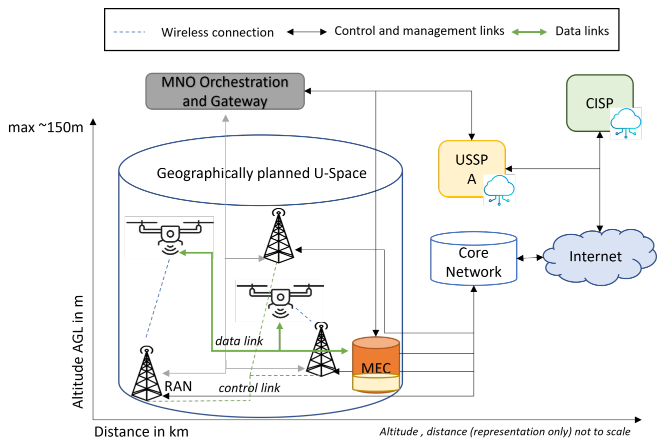

Figure 3. Communication infrastructure in a USSP-centric de-centralised model

places MEC infrastructure under the control of the CISP as a neutral and trusted platform. Here, UAS traffic connects through the RAN and MNO into the CISP-managed MEC through specific control and data links. Tactical decisioning may run as CISP-native services or as lightweight USSP containers hosted on the MEC, represented through green disc (an application service of USSP A). This is carried out with the CISP ensuring schema compliance, SLA enforcement, and consistent behavior.

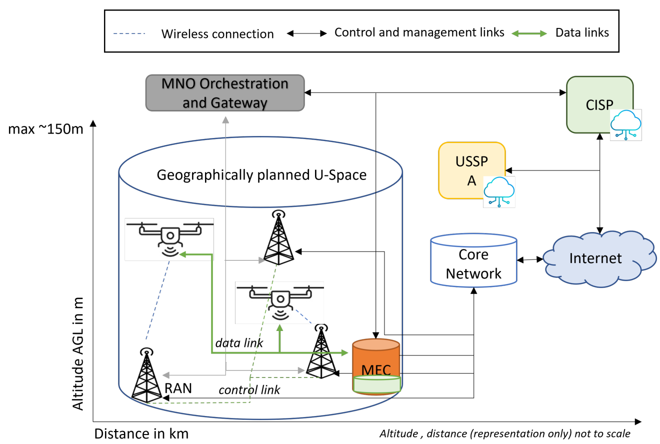

Figure 4. Communication infrastructure in a CISP-centric de-centralised model

TABLE I. 3GPP - 5G PROFILE LATENCY BUDGET USED IN SIMULATION

Components	Centralised	USSP-centric MEC	CISP-centric MEC
RAN	5–10 ms	5–10 ms	5–10 ms
Round-Trip Time	20-40 ms	5–10 ms	5–10 ms
MEC processing time	0 ms	20-40 ms	10-20 ms
Handover time	0 ms	20-40 ms	20-40 ms

In this model (figure 4), USSPs subscribe to MEC event topics or deploy tactical functions locally, while their own clouds remain responsible for strategic services such as billing, storage, or long-term planning. Session continuity during U-Space handovers is simplified, as the USSP- centric MEC or CISP-centric MEC replicates state across its MEC nodes (based on MNO subscription), ensuring uninterrupted control-plane and user-plane flows. This approach enhances consistency, interoperability, and regulatory compliance by anchoring tactical data exchange in a single neutral layer.

Thus, by aligning data mesh principles with MEC deployments, U-Space actors can achieve both autonomy and interoperability: preserving local ownership of critical data while maintaining the shared situational awareness essential for safety and compliance. The models described above - centralised, USSP-centric MEC, and CISP-centric MEC have been further evaluated in a preliminary simulation study, focusing on their latency, scalability, and continuity characteristics. These results are discussed in the following section.

IV. Case Study Analysis and Evaluation

To evaluate the feasibility of the proposed architectures, we modeled three scenarios : centralised cloud-only, USSP-centric MEC, and CISP-hosted MEC and measured their impact on latency, scalability, and continuity under increasing UAS density and multiple USSP configurations. We assume standard 3GPP procedures for control plane session continuity and mobility for control and management link, and focus on user plane (data link) [34] latency and application migration, particularly how MEC placement and federation affect service responsiveness.

The preliminary results presented here focus on the feasibility of the aforementioned models for UAS traffic management in U-Space. The simplified simulation environment consists U-Space and the operation area is covered by five 5G RAN. The assumptions on latency budget in table I are based 3GPP and ETSI framework. The U-Space consists of up to ten USSPs and up to ten UAS per USSP, entering the U-Space - randomised in space and time. The one - central CISP governs critical decisions of this simulated U-Space. The messages between these entities are generated telemetry, flight authorization, or de-confliction.

In the centralised model, where MEC is absent and services run only in cloud infrastructures at CISP, the mean latency of above 100 ms shows a clear upward trend as the number of UASs increases. This behavior stems from the requirement that all telemetry and authorization traffic traverse the full backhaul to centralised clouds, with no local offload at the edge.

While the continuity of connections is preserved, the latency observed renders this architecture unsuitable for high-density U-Space operations where sub-50 ms responsiveness is required [35]. The USSP-centric MEC model demonstrates clear latency improvements in our simulations in figure 5.Here, user plane offloading happens directly at the regional or metro edge, ensuring tactical decisioning occurs close to the UAS. The plots confirm consistently lower latency per UAS compared to the centralised setup, but they also reveal instability when the number of USSPs increases beyond three to five. Each USSP must provision its own MEC containers, and inter-USSP handovers create latency spikes during MEC-to-MEC migration as in figure 6.

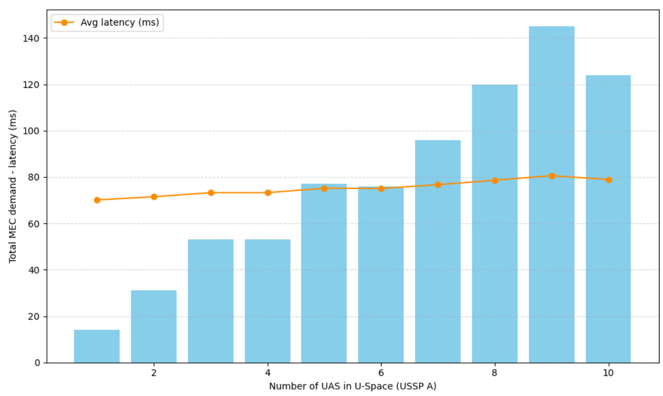

Figure 5. Latency and average latency (in millisecond) of USSP-MEC with increasing UAS in the same U-Space

The figure 6 illustrates the latency experienced by a UAS during a handover from one U-Space MEC to another. In the initial phase (0–20 s), the UAS is connected to the MEC inthe current U-Space, maintaining a stable latency of around 15 ms. Between 20–30 s, the handover occurs: latency spikes to 50 ms due to session relocation, re-anchoring, and application state migration across MECs. After the handover (30–60 s), the UAS anchors to the MEC in neighboring U-Space, restoring stable latency around 18 ms. This highlights that MEC handovers introduce short disruptions but capable to handle end-to-end latency far below the levels expected in centralised CISP Cloud-only model, demonstrating the benefit of local edge compute for tactical U-Space operations. Furthermore, Scalability is thus constrained by the operational overhead. In extreme latency cases, latency values cluster between 138–144 ms with peaks around 142 ms, and some drones reaching 150 ms, showing clear variability and jitter. This wider distribution reflects fragmentation, as each USSP manages its own edge resources, leading to higher average latency and inconsistent service quality. The CISP-centric MEC model provides the most balanced results, as seen in figure 7. This plot shows stable latency around 15–17 ms, with only minor increases as more USSPs and drones are added. This demonstrates that a shared neutral MEC at the CISP level scales more consistently, maintaining low latency and reliable continuity even under growing multi-USSP load. By deploying MEC nodes as neutral, metro-edge platform controlled by the CISP can broker tactical loops consistently across all USSPs. Here, user plane offloading occurs at the CISP managed

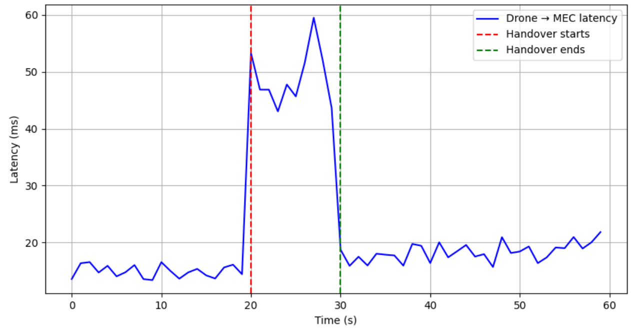

Figure 6. Latency (in millisecond) of a UAS during USSP-centric MEC service migration from one U-Space to neighboring U-Space

MEC with session continuity ensured by standard relocation procedures across U-Spaces. The latency plots show stable, low-variance performance even as drone and USSP numbers scale, demonstrating the efficiency of shared infrastructure and neutral orchestration. From a control and management link standpoint, UAS maintain continuity as they cross U-Spaces since the CISP replicates state between its MEC nodes, minimizing disruptions during inter-MNO handovers. The shared event bus model ensures that USSP applications can still run on the MEC while retaining their cloud backends for strategic services. Figure 7 highlights how latency remains consistently within acceptable bounds, unlike the divergence observed in USSP-centric decentralised MEC model.

TABLE II. LATENCY CHARACTERISTICS IN THE THREE 5G EDGE COMPUTING MODELS

Model	Latency Profile	Key Characteristics
Centralised	High (40-80 ms)	grows with load,Stable continuity, single point of failure
USSP-MEC	Low (<20 ms) but spikes during migration	Good for single and multi-USSPs,high governance required for scalability
CISP-MEC	Consistent low latency (10-20 ms)	Best balance, smooth handovers possible, easily scalable

This figure 8 compares mean latency across the three proposed architectures: centralised cloud-only, USSP-centric MEC, and CISP-centric MEC as the number of USSPs increases in the simulated environment. The MEC-enabled models significantly outperform the cloud-only approach in latency, particularly in dense multi-USSP settings as in comparision table II.

The CISP-centric MEC model exhibits the lowest latency, emphasizing its scalability and suitability for large-scale, safety-critical drone traffic management. However, the main trade-off here is governance: CISP becomes the critical operational anchor, raising questions of trust, liability, and neutrality. Nevertheless, technically this approach aligns best with 3GPP MEC federation goals and offers superior scalability and continuity guarantees.

TABLE III. QUALITATIVE KPI COMPARISON ACROSS ARCHITECTURES

KPI	Centralised	USSP-centric MEC	CISP-hosted MEC
Latency	High (40-80 ms, scales poorly)	Low (<20 ms) but jitter under handovers	Stable low (10-20 ms) across scales
Scalability	Limited - cloud congestion	MEC proliferation per USSP	Best - shared neutral MEC per U-Space
Continuity	Control plane only	Better- handover migration issues	Best - CISP replicates state across MECs
QoS Control	Limited	Better (per-USSP tuning possible)	Best via neutral orchestration
Governance	Best - simple (per-cloud)	Limited -Fragmented	Better- Neutral

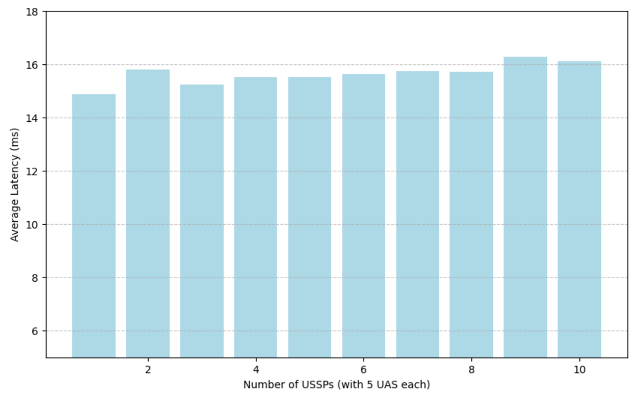

Figure 7. CISP-centric MEC instances and corresponding average latency (in milliseconds) in the U-Space

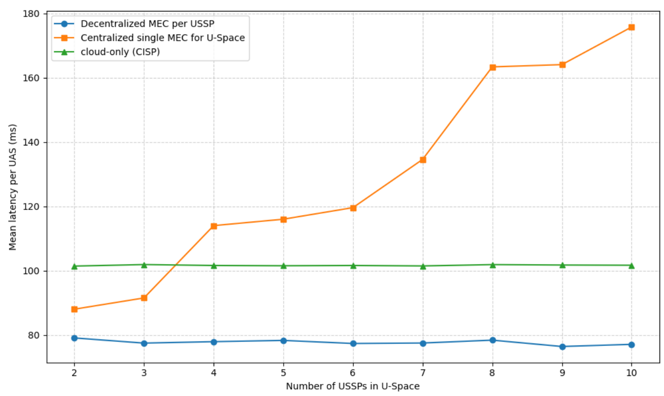

Figure 8. Comparison of mean latency(in milliseconds) per UAS with increasing number of USSPs for three proposed models in simulated U-Space

V. DISCUSSION AND CONCLUSION

This paper proposes a MEC-enabled, data-mesh-driven architecture for U-Space traffic management, addressing challenges of latency, scalability, and federated governance across heterogeneous UAS operations. The three considered architectures centralised cloud-only, USSP-centric MEC, and CISP-centric MEC each fit specific U-Space scales and operational complexities. Centralised solutions suit smaller U-Spaces with lighter tactical demands, whereas metro or regional-edge MEC deployments optimise latency, scalability, and multi-stakeholder interoperability for dense, complex environments.

Inter-MEC communication is fundamental for session continuity, coordinated tactical decision-making, and seamless service migration as UAVs move between U-Spaces. However, federating MECs across stakeholders introduces complex challenges such as identity management, synchronization, and consistent QoS enforcement. To compare architectures more systematically, table III summarises the qualitative KPI trade-offs derived from our simulations and analysis.

In this context, the adoption of data mesh principles provides a governance framework that balances autonomy with interoperability. By applying domain-oriented ownership, USSPs, CISPs, and UAS operators retain responsibility for their respective data products; through data-as-a-product, information such as telemetry streams, flight intents, and geo-awareness constraints is standardised and consumable across domains; self-serve infrastructure enables stakeholders to publish and subscribe via shared MEC and cloud resources without bespoke integrations; and federated governance enforces schemas, SLAs, and security policies across all actors. This ensures that local tactical autonomy is preserved while maintaining the global situational awareness needed for regulatory compliance and operational safety.

Each architecture exhibits weaknesses. The centralised CISP-edge model concentrates control, posing political risks, creating single points of failure, and raising liability concerns about data exposure and commercial negotiations over cost-sharing and resource allocation. The USSP-centric model aligns better with commercial interests but risks fragmentation, interoperability conflicts, and scaling inefficiencies, especially if many USSPs coexist. Both face common technical issues, including maintaining sub-50 ms handover continuity under MEC federation immaturity, difficulties enforcing QoS at UAV altitudes, and managing semantic inconsistencies across USSPs [30]. Failure modes vary: USSP-centric MEC risks mission drops when MEC coverage gaps exist, while CISP-centric MEC is vulnerable to widespread outages or false alert propagation during failures.

Scalability and cost trade-offs are core operational considerations. Expanding MEC coverage improves resilience but increases deployment and operational expenses. The CISP-edge approach appears more practical for safety and regulation if political and trust challenges are overcome; otherwise, market forces may favor USSP-centric fragmentation with potential safety implications.

Furthermore, ongoing and future work will focus on developing a comprehensive simulation framework that incorporates geographically distributed regional and metro MEC deployments to capture realistic latency behavior and message-level performance metrics. The framework will evaluate traffic prioritization schemes and event-driven communication mechanisms across the proposed architectures to assess their responsiveness, robustness, and scalability under varying load conditions. These studies are expected to yield actionable insights for selecting the most suitable architectural model based on U-Space scale and operational profiles, ultimately supporting secure, resilient, and cost-effective UAS traffic management deployments.

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